

Advanced Statistical Learning

Linear Methods for Regression

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Outline

Intro

Linear Regression & Least Squares

Subset Selection

Shrinkage methods

Methods Using Derived Input Directions

Advantages of Linear Models

- ▶ Simple and often provide an adequate and interpretable description of how the inputs affect the output.
- ▶ Sometimes outperform fancier nonlinear models, especially in situations with small numbers of training cases, low signal-to-noise ratio or sparse data.
- ▶ Can be applied to transformations of the inputs and this considerably expands their scope.

Recap of Linear Regression

$$f(X) = \beta_0 + \sum_{j=1}^P X_j \beta_j$$

- ▶ β_j unknown parameter
- ▶ Sources of X_j
 - ▶ quantitative inputs
 - ▶ transformations
 - ▶ basis expansions
 - ▶ dummy coding of qualitative variables
 - ▶ interactions between variables
- ▶ Model is linear in β

Estimation

Training data:

$$(x_1, y_1), \dots, (x_N, y_N)$$

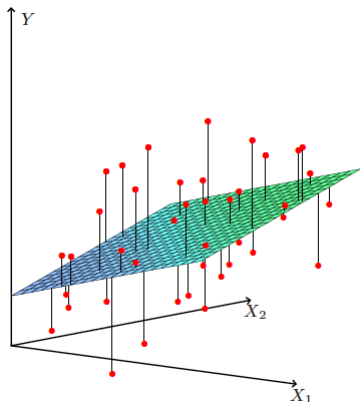
Feature measurements for the i_{th} case:

$$x_i = (x_{i1}, x_{i2}, \dots, x_{ip})^T:$$

Least squares:

choose $\beta = (\beta_0, \beta_1, \dots, \beta_p)^T$ to minimize the Residual Sum of Squares.

$$\blacktriangleright RSS = \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2$$



Matrix notation

- ▶ X : $N \times (p + 1)$ matrix
- ▶ Y : N vector

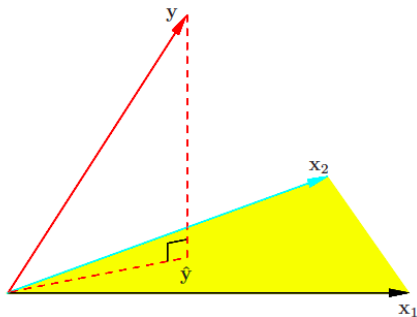
$$RSS(\beta) = (y - X\beta)^T (y - X\beta)$$

- ▶ Quadratic function of β
- ▶ Differentiating w.r.t. β
 - ▶ $\frac{\partial RSS}{\partial \beta} = -2X^T (y - X\beta)$
 - ▶ $\frac{\partial^2 RSS}{\partial \beta \partial \beta^T} = 2X^T X$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

- ▶ Assume X has full column rank
- ▶ Fitted value $\hat{y} = X\hat{\beta} = X(X^T X)^{-1} X^T y$
- ▶ Hat matrix $H = X(X^T X)^{-1} X^T$

Geometry in $\mathbb{R}^N$



Minimize $RSS(\beta) = ||y - x\beta||^2$ by choosing $\hat{\beta}$ such that the residual vector $y - \hat{y}$ is orthogonal to the subspace spanned by X .

$\hat{y}$ is the orthogonal projection of y and H is called projection matrix.

Sampling Properties of $\hat{\beta}$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

- ▶ Without any assumptions, $\text{Var}(\hat{\beta}) = (X^T X)^{-1} \sigma^2$
- ▶ $\hat{\sigma}^2 = \frac{1}{N-p-1} \sum_{i=1}^N (y_i - \hat{y}_i)^2$

Additional assumptions

- ▶ $y = \beta_0 + \sum_{j=1}^p X_j \beta_j + \epsilon$
- ▶ $\epsilon \sim \mathcal{N}(0, \sigma^2)$

Sampling distribution of $\hat{\beta}$, $\hat{\sigma}^2$

- ▶ $\hat{\beta} \sim \mathcal{N}(\beta, (X^T X)^{-1} \sigma^2)$
- ▶ $(N - p - 1) \hat{\sigma}^2 \sim \sigma^2 \chi_{N-p-1}^2$

Hypothesis Testing - single parameter

When σ^2 is known (rare)

- ▶ v_j is the j_{th} diagonal element of $(X^T X)^{-1}$
- ▶ $z_j = \frac{\hat{\beta}_j}{\sigma \sqrt{v_j}}$
- ▶ Under H_0 , $z_j \sim \mathcal{N}(0, 1)$

When σ^2 is unknown (most common)

- ▶ $z_j = \frac{\hat{\beta}_j}{\hat{\sigma} \sqrt{v_j}}$
- ▶ Under H_0 , $z_j \sim t(N - p - 1)$

The differences between $t(N - p - 1)$ and standard normal become negligible as the sample size increases

Hypothesis Testing - group of coefficients

Bigger model

- ▶ residual sum-of-squares for the least squares fit RSS_1
- ▶ $p_1 + 1$ parameters

Nested smaller model

- ▶ residual sum-of-squares for the least squares fit RSS_0
- ▶ $p_0 + 1$ parameters

$$F = \frac{(RSS_0 - RSS_1)/(P_1 - P_0)}{RSS_1/(N - P_1 - 1)}$$

- ▶ Measures the change in RSS per additional parameter in the bigger model
- ▶ Under H_0 , $F \sim \mathcal{F}_{p_1 - p_0, N - p_1 - 1}$

Confidence Interval / Region

β_j : $1 - 2\alpha$ confidence interval

► $\left(\hat{\beta}_j - z^{1-\alpha} v_j^{1/2} \hat{\sigma}, \hat{\beta}_j + z^{1-\alpha} v_j^{1/2} \hat{\sigma} \right)$

Entire parameter vector β : $1 - \alpha$ confidence region

► $\mathcal{C}_\beta = \left\{ \beta \mid \left(\hat{\beta} - \beta \right)^\top X^\top X \left(\hat{\beta} - \beta \right) \leq \hat{\sigma}^2 \chi_{p+1}^2{}^{1-\alpha} \right\}$

Gauss Markov Theorem

Why least squares?

Gauss Markov Theorem: least squares estimates of the parameters β have the smallest variance among all linear unbiased estimates

In mathematical term: suppose $\theta = \alpha^\top \beta$

- ▶ $\hat{\theta} = \alpha^\top \hat{\beta}$ is unbiased for θ
- ▶ Consider any other linear estimator $\tilde{\theta} = c^\top y$ such that $E(c^\top y) = \alpha^\top \beta$
- ▶ According to Gauss Markov Theorem $Var(\alpha^\top \hat{\beta}) < Var(c^\top y)$

Gauss Markov Theorem - Proof

Gauss Markov Theorem

Mean squared error of an estimator $\tilde{\theta}$ in estimating θ :

$$MSE(\tilde{\theta}) = E(\tilde{\theta} - \theta)^2 = E(\tilde{\theta} - E(\tilde{\theta}) + E(\tilde{\theta}) - \theta)^2 = Var(\tilde{\theta}) + Bias(\tilde{\theta})^2$$

Gauss-Markov theorem: least squares estimator has the smallest mean squared error of all linear estimators with no bias.

How about biased estimator (best subset, shrinkage)?

- ▶ Might have smaller MSE by trading a little bias for a larger reduction in variance.

"All models are wrong, some are useful"

- ▶ Picking the right model amounts to creating the right balance between bias and variance.

Gauss Markov Theorem

Mean squared error is intimately related to prediction accuracy:
consider prediction of the new response at input x_0

$$Y_0 = f(x_0) + \epsilon_0$$

Then the expected prediction error of an estimate $\tilde{f}(x_0) = x_0^T \tilde{\beta}$ is

$$\begin{aligned} E(Y_0 - \tilde{f}(x_0))^2 &= \sigma^2 + E(x_0^T \tilde{\beta} - f(x_0))^2 \\ &= \sigma^2 + MSE(\tilde{f}(x_0)) \end{aligned}$$

Therefore, expected prediction error and mean squared error differ only by the constant σ^2 , representing the variance of the new observation y_0

Multiple Linear Regression

$$y_i = \beta_0 + \sum_{j=1}^p x_{ij}\beta_j \text{ where } p > 1$$

When $x_1, x_2, \dots, x_p$ are orthogonal ($\langle x_j, x_k \rangle = 0$ for all $j \neq k$)

▶ $\hat{\beta}_j = \frac{\langle x_j, y \rangle}{\langle x_j, x_j \rangle}$, same as univariate solution

When $x_1, x_2, \dots, x_p$ are not orthogonal

- ▶ initialize $z_0 = x_0 = 1$
- ▶ for $j = 1, \dots, p$
 - ▶ regress x_j on $z_0, \dots, z_{j-1}$ to produce coefficients
 $\hat{\gamma}_{lj} = \langle z_l, x_j \rangle / \langle z_l, z_l \rangle$, $l = 0, \dots, j-1$ and residual vector
 $z_j = x_j - \sum_{k=0}^{j-1} \hat{\gamma}_{kj} z_k$
- ▶ Regress y on the residual z_p to give the estimate $\hat{\beta}_p$

Also known as Gram-Schmidt procedure. Note: if x_j is highly correlated with x_k , z_j will be close to zero, and β_j will be very unstable.

Multiple Outputs $Y_1, Y_2, \dots, Y_K$

$$Y_k = \beta_{0k} + \sum_{j=1}^P X_j \beta_{jk} + \epsilon_k = f_k(X) + \epsilon_k$$

or in matrix notation $Y = XB + E$

If errors $\epsilon = (\epsilon_1, \epsilon_2, \dots, \epsilon_k)$ are i.i.d.

- ▶ $RSS(B) = \sum_{k=1}^K \sum_{i=1}^N (y_{ik} - f_k(x_i))^2 = \text{tr}[(Y - XB)^T(Y - XB)]$
- ▶ $\hat{B} = \text{argmin}_B RSS(B) = (X^T X)^{-1} X^T Y$
- ▶ Same form as univariate output

If errors $\epsilon = (\epsilon_1, \epsilon_2, \dots, \epsilon_k)$ are correlated

- ▶ $RSS(B, \Sigma) = \sum_{i=1}^N (y_i - f(x_i))^T \Sigma^{-1} (y_i - f(x_i))$
- ▶ $\hat{B} = \text{argmin}_B RSS(B, \Sigma) = (X^T X)^{-1} X^T Y$
- ▶ Note: If Σ_i vary among observations, results no longer hold

Satisfied with Least Squares Estimates?

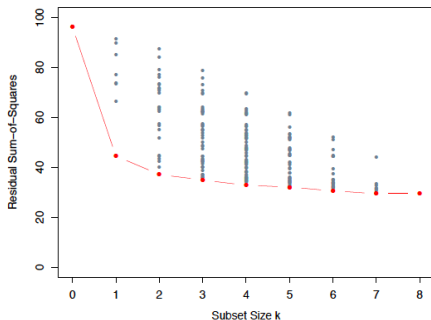
Prediction accuracy

- ▶ least squares estimates often have low bias but large variance.
- ▶ Prediction accuracy can sometimes be improved by shrinking or setting some coefficients to zero.

Interpretability

- ▶ Determine a smaller subset that exhibit the strongest effects.
- ▶ Sacrifice some of the small details in order to get the "big picture"

Best-Subset Selection



Select k ?

- ▶ Cross validation
- ▶ AIC
- ▶ ...

For each $k \in \{0, 1, 2, \dots, p\}$, find the subset of size k that gives smallest residual sum of squares.

- ▶ OK when k is small (leaps and bounds algorithm).
- ▶ Intractable as k becomes large (> 40)

Forward, Backward, Stepwise Selection

Forward

- ▶ Starts with the intercept
- ▶ Sequentially adds the predictor that most improves the fit
- ▶ Must determine model size k

Backward

- ▶ Starts with the full model
- ▶ Sequentially deletes the predictor that has the least impact on fit
- ▶ Candidate for dropping is the variable with the smallest Z-score

Stepwise

- ▶ Consider both forward and backward moves at each step
- ▶ Select the “best” of the two
- ▶ R step function: at each step add or drop based on AIC

Satisfied with Sub-set Selection?

Pros

- ▶ Interpretable
- ▶ Possibly lower prediction error than the full model.

Cons

- ▶ Discrete process - variables either retained or discarded
- ▶ Often exhibits high variance

Ridge Regression

Shrinks the regression coefficients by imposing a penalty on their size.

$$\hat{\beta}^{ridge} = \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}$$

which is equivalent to

$$\hat{\beta}^{ridge} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 \text{ subject to } \sum_{j=1}^p \beta_j^2 \leq t$$

- ▶ β_0 is left out of the penalty term
- ▶ Normalize inputs before solving for solutions
- ▶ In matrix notation $RSS(\lambda) = (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta$
- ▶ $\hat{\beta}^{ridge} = (X^T X + \lambda I)^{-1} X^T y$

Ridge Regression

$$\hat{\beta}^{ridge} = (X^T X + \lambda I)^{-1} X^T y$$

- ▶ $\hat{\beta}^{ridge}$ is again a linear function of y
- ▶ Makes it nonsingular even if $X^T X$ is not of full rank
- ▶ When X are orthogonal, $\hat{\beta}^{ridge}$ are just a scaled version of $\hat{\beta}^{OLS}$
- ▶ Equivalent Bayesian context - Gaussian prior for β

Ridge Regression

Consider SVD of the centered input matrix $\mathbf{X}$: $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T$

- ▶ $\mathbf{U}^{n \times p}$ orthogonal, $\mathbf{D}^{p \times p}$ diagonal, $\mathbf{V}^{p \times p}$ orthogonal

$$\mathbf{X}\hat{\beta}^{OLS} = \mathbf{U}\mathbf{U}^T \mathbf{y}$$

$$\mathbf{X}\hat{\beta}^{ridge} = \mathbf{U}\mathbf{D}(\mathbf{D}^2 + \lambda \mathbf{I})^{-1} \mathbf{D}\mathbf{U}^T \mathbf{y} = \sum_{j=1}^p \mu_j \frac{d_j^2}{d_j^2 + \lambda} \mu_j^T \mathbf{y}$$

- ▶ u_j are the columns of $\mathbf{U}$
- ▶ d_j are the singular values of $\mathbf{X}$
- ▶ Greater amount of shrinkage is applied to the coordinates of basis vectors with smaller d_j
- ▶ effective degrees of freedom $df(\lambda) = \sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda}$

Ridge Regression

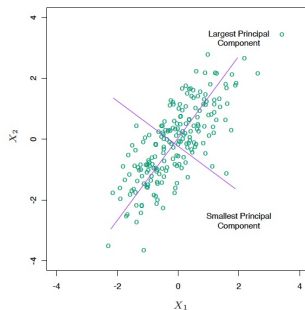
What does small values of d_j^2 mean?

$$X^T X = V D^2 V^T$$

- ▶ d_j^2 are the eigen values of $X^T X$
- ▶ eigen vectors v_j are the principal component directions of X

Small singular values d_j correspond to directions in the column space of X having small variance, and ridge regression shrinks these directions the most

Ridge Regression



Assume response tend to vary most in the directions of high variance of the inputs

- ▶ reasonable since predictors are often chosen since they vary with response
- ▶ but need not hold in general

The LASSO

L_2 penalty $\sum_{j=1}^p \beta_j^2$ is replaced by L_1 penalty $\sum_{j=1}^p |\beta_j|$.

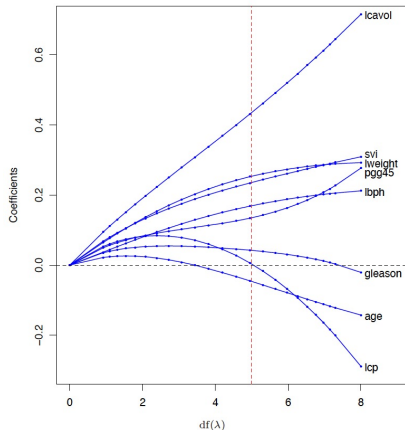
$$\hat{\beta}^{lasso} = \underset{\beta}{\operatorname{argmin}} \left\{ \frac{1}{2} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

Which is equivalent to

$$\hat{\beta}^{lasso} = \underset{\beta}{\operatorname{argmin}} \frac{1}{2} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 \text{ subject to } \sum_{j=1}^p |\beta_j| \leq t$$

- ▶ Non-linear in y
- ▶ No closed form expression
- ▶ $\hat{\beta}^{lasso}$ becomes $\hat{\beta}^{OLS}$ when $t > \sum_{j=1}^p |\beta_j^{OLS}|$
- ▶ $\hat{\beta}^{lasso} = 0$ when $t \rightarrow 0$
- ▶ Continuous subset selection.

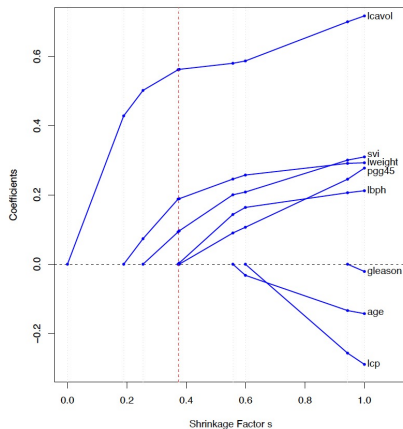
The Ridge solution path



Effective degrees of freedom $df(\lambda) = \sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda}$

- ▶ Smooth solution path
- ▶ vertical line (optimal value) chosen by cross-validation

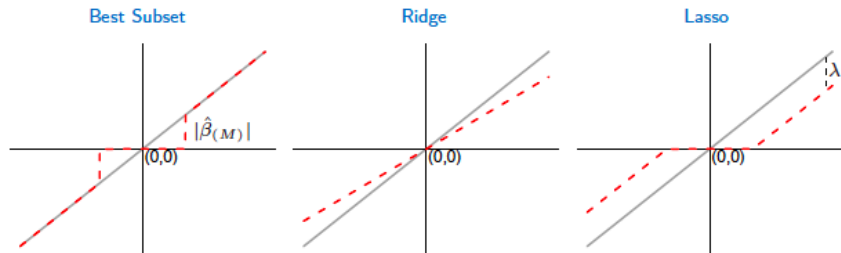
The LASSO solution path



Shrinkage factor $s = t / \sum_1^p |\hat{\beta}_j|$

- ▶ lasso profiles hit 0, while those for ridge do not
- ▶ lasso profiles are piecewise linear

Comparisons - Orthonormal Inputs



Best Subset

- ▶ Drop all variables with coefficients smaller than the M-th
- ▶ “hard-thresholding.”

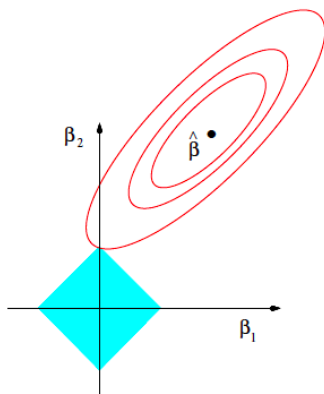
Ridge

- ▶ Proportional shrinkage

LASSO

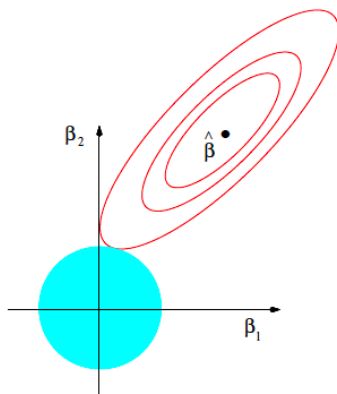
- ▶ translates by a constant factor λ , truncating at zero.
- ▶ “soft thresholding”

Comparisons - Nonorthogonal Inputs



LASSO

$$|\beta_1| + |\beta_2| \leq t$$

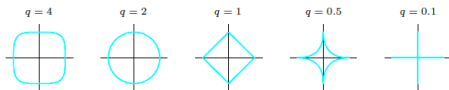


Ridge

$$\beta_1^2 + \beta_2^2 \leq t^2$$

Generalization of Ridge and LASSO

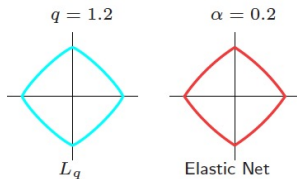
$$\tilde{\beta} = \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p |\beta_j|^q \right\}, \quad q \geq 0$$



- ▶ $q = 0$, subset selection
- ▶ $0 \leq q < 1$, non-convex constraint regions, difficult optimization
- ▶ $q = 1$, LASSO
- ▶ $1 < q < 2$: "compromise" between Ridge and LASSO, but not desirable
- ▶ $q = 2$, ridge
- ▶ Learn optimal q from data? not worth the efforts

Generalization of Ridge and LASSO

Elastic net: $\lambda \sum_{j=1}^p \left(\alpha \beta_j^2 + (1 - \alpha) |\beta_j| \right)$, $0 \leq \alpha \leq 1$



- ▶ visually similar, but elastic net has sharp (non-differentiable) corners
- ▶ $\sum_j |\beta_j|^q$ penalty not recommend
- ▶ Elastic net selects variables like lasso, and shrinks coefficients of correlated variables like ridge

Least Angle Regression

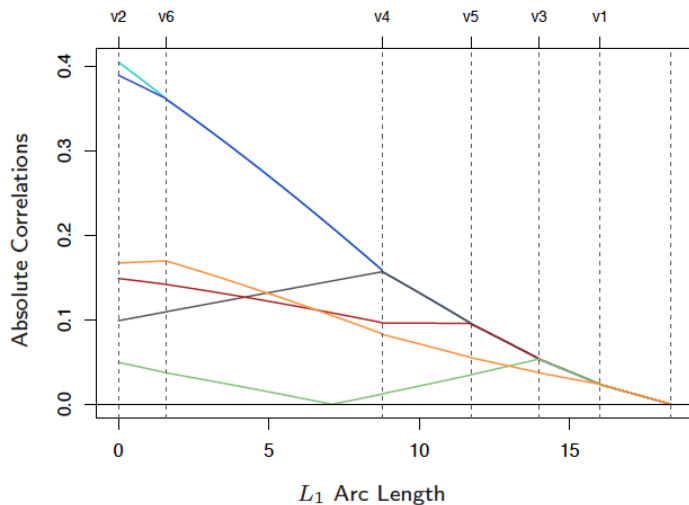
Similar to forward stepwise, but only enters “as much” of a predictor as it deserves.

- ▶ Identifies the variable most correlated with the response
- ▶ Moves the coef continuously toward its least-squares value
- ▶ As soon as another variable “catches up”, stop
- ▶ The second variable joins the active set
- ▶ Repeat...

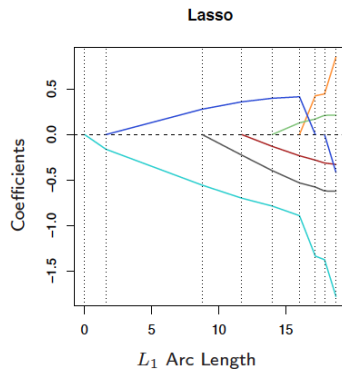
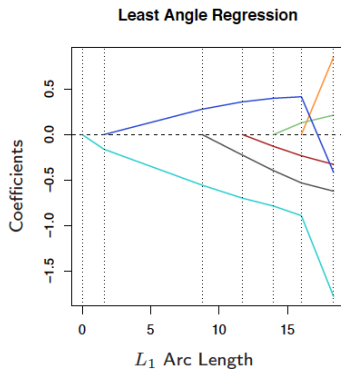
Least Angle Regression - Algorithm

- ▶ Standardize the predictors to have mean zero and unit norm. Start with the residual $r = y - \bar{y}$, $\beta_1, \beta_2, \dots, \beta_p = 0$.
- ▶ Find the predictor x_j most correlated with r
- ▶ Move β_j from 0 towards its least-squares coefficient $\langle x_j, r \rangle$, until some other competitor x_k has as much correlation with the current residual as does x_j
- ▶ Move β_j and β_k in the direction defined by their joint least squares coefficient of the current residual on (x_j, x_k) until some other competitor x_l has as much correlation with the current residual
- ▶ Continue in this way until all p predictors have been entered. After $\min(N - 1, p)$ steps, we arrive at the full least-squares solution.

Least Angle Regression - Progression



Least Angle Regression & LASSO



Methods Using Derived Input Directions

Some Algebra

Eigen decomposition (square matrix)

- ▶ A : eigenvalues $\lambda_1, \dots, \lambda_k$, eigenvectors $\nu_1, \dots, \nu_k$
- ▶ Let $D \equiv \begin{bmatrix} \lambda_1, 0, \dots, 0 \\ 0, \lambda_2, \dots, 0 \\ \dots, \dots, \dots, \dots \\ 0, 0, \dots, \lambda_k \end{bmatrix}$, $P \equiv [\nu_1, \nu_2, \dots, \nu_k]$
- ▶ $A = PDP^{-1}$, $A^2 = PD^2P^{-1}$, ..., $A^n = PD^nP^{-1}$

Singular value decomposition (any matrix $m \times n$)

- ▶ $A = U\Sigma V^T$, where $U_{m \times n}$, $\Sigma_{n \times n}$ and $V_{n \times n}$
- ▶ U is orthonormal and span the column space of A ,
- ▶ Σ is diagonal with singular values of A
- ▶ V has eigen vectors of $A^T A$ in the column

Singular Value Decomposition

Any $m \times n$ real matrix: $A \in \mathbb{R}^{m \times n}$

$$A = U \Sigma V^T$$

- ▶ $U \in \mathbb{R}^{m \times m}$ orthonormal, $UU^T = I$
- ▶ $V \in \mathbb{R}^{n \times n}$ orthonormal, $VV^T = I$
- ▶ $\Sigma \in \mathbb{R}^{m \times n}$ rectangular diagonal, $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_p)$ with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p \geq 0$, $p = \min(m, n)$
- ▶ (U : left singular vector, V : right singular vector, σ 's: singular values)

Singular Value Decomposition - Example

$$A = \begin{Bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{Bmatrix}$$

$$\blacktriangleright U = \begin{Bmatrix} 0 & 0 & \sqrt{0.2} & 0 & \sqrt{0.8} \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \sqrt{0.8} & 0 & -\sqrt{0.2} \end{Bmatrix}, \Sigma = \begin{Bmatrix} 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{5} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{Bmatrix}, V^T = \begin{Bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{Bmatrix}$$

Singular Value Decomposition - Example Cont'd

$$A = \begin{Bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{Bmatrix}$$

$$\blacktriangleright U = \begin{Bmatrix} 0 & 0 & \sqrt{0.2} & \sqrt{0.4} & -\sqrt{0.4} \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{0.5} & \sqrt{0.5} \\ 0 & 0 & \sqrt{0.8} & -\sqrt{0.1} & \sqrt{0.1} \end{Bmatrix}, \Sigma = \begin{Bmatrix} 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{5} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{Bmatrix}, V^T = \begin{Bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{Bmatrix}$$

Existence of SVD

Theorem:

If $A \in \mathbb{R}^{m \times n}$ is a real matrix, $\exists$ orthonormal matrix $U \in \mathbb{R}^{m \times m}$, orthonormal matrix $V \in \mathbb{R}^{n \times n}$ and rectangular diagonal matrix $\Sigma \in \mathbb{R}^{m \times n}$ with descending and non-negative diagonal elements, such that

$$A = U\Sigma V^T$$

► Proof is not required

Compact SVD

If $A \in \mathbb{R}^{m \times n}$ is a real matrix with $\text{rank}(A) = r \leq \min(m, n)$, compact SVD:

$$A = U_r \Sigma_r V_r^T$$

- ▶ $U_r \in \mathbb{R}^{m \times r}$: first r columns of $U \in \mathbb{R}^{m \times m}$
- ▶ $V_r \in \mathbb{R}^{n \times r}$: first r columns of $V \in \mathbb{R}^{n \times n}$
- ▶ $\Sigma_r \in \mathbb{R}^{r \times r}$: first r diagonal elements of $\Sigma \in \mathbb{R}^{m \times n}$
- ▶ $\text{rank}(\Sigma_r) = \text{rank}(A)$

Compact SVD - Example

$$A = \begin{Bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{Bmatrix}, \quad \text{rank}(A) = 3$$

$$\blacktriangleright U = \begin{Bmatrix} 0 & 0 & \sqrt{0.2} & 0 & \sqrt{0.8} \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \sqrt{0.8} & 0 & -\sqrt{0.2} \end{Bmatrix}, \quad \Sigma = \begin{Bmatrix} 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{5} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{Bmatrix}, \quad V^T = \begin{Bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{Bmatrix}$$

$$\blacktriangleright U_3 = \begin{Bmatrix} 0 & 0 & \sqrt{0.2} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sqrt{0.8} \end{Bmatrix}, \quad \Sigma_3 = \begin{Bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \sqrt{5} \end{Bmatrix}, \quad V_3^T = \begin{Bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{Bmatrix}$$

Truncated SVD

If $A \in \mathbb{R}^{m \times n}$ is a real matrix with $\text{rank}(A) = r \leq \min(m, n)$, $0 < k < r$, truncated SVD:

$$A \approx U_k \Sigma_k V_k^T$$

- ▶ $U_k \in \mathbb{R}^{m \times k}$: first k columns of $U \in \mathbb{R}^{m \times m}$
- ▶ $V_k \in \mathbb{R}^{n \times k}$: first k columns of $V \in \mathbb{R}^{n \times n}$
- ▶ $\Sigma_k \in \mathbb{R}^{k \times k}$: first k diagonal elements of $\Sigma \in \mathbb{R}^{m \times n}$
- ▶ $\text{rank}(\Sigma_k) < \text{rank}(A)$

Truncated SVD - Example

$$A = \begin{Bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{Bmatrix}, \quad \text{rank}(A) = 3, k = 2$$

$$\blacktriangleright U_2 = \begin{Bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{Bmatrix}, \Sigma_2 = \begin{Bmatrix} 4 & 0 \\ 0 & 3 \end{Bmatrix}, V_2^T = \begin{Bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{Bmatrix}$$

$$\blacktriangleright A_2 = \begin{Bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{Bmatrix}$$

Geometry of SVD

$A \in \mathbb{R}^{m \times n}$: a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$

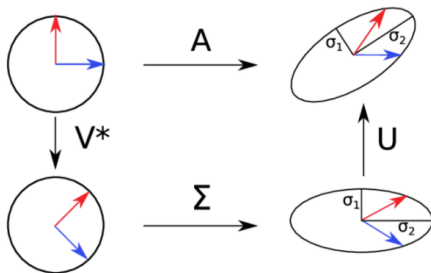
$$T : x \rightarrow Ax \quad (U\Sigma V^T x)$$

- ▶ $x \in \mathbb{R}^n$, $Ax \in \mathbb{R}^m$
- ▶ A linear transformation can be decomposed into 3 steps: rotation, stretch/scale, rotation again
 - ▶ coordinates rotated by V^T (orthonormal basis in $\mathbb{R}^n$)
 - ▶ coordinates stretched or scaled by Σ ($\sigma_1, \sigma_2, \dots, \sigma_n$)
 - ▶ coordinates rotated by U (orthonormal basis in $\mathbb{R}^m$)

Geometry of SVD

$A \in \mathbb{R}^{m \times n}$: a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$

$$T: x \rightarrow Ax \quad (U \Sigma V^T x)$$



$$A = U \Sigma V^*$$

Geometry of SVD - Example

$$A = \begin{Bmatrix} 3 & 1 \\ 2 & 1 \end{Bmatrix}$$

$$U = \begin{Bmatrix} 0.8174 & -0.5760 \\ 0.5760 & 0.8174 \end{Bmatrix}, \Sigma = \begin{Bmatrix} 3.8643 & 0 \\ 0 & 0.2588 \end{Bmatrix}, V^T = \begin{Bmatrix} 0.9327 & 0.3606 \\ -0.3606 & 0.9327 \end{Bmatrix}$$

Consider an orthonormal basis in $\mathbb{R}^2$: $e_1 = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$, $e_2 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$

► rotation by V^T : $V^T e_1 = \begin{Bmatrix} 0.9327 \\ -0.3606 \end{Bmatrix}$, $V^T e_2 = \begin{Bmatrix} 0.3606 \\ 0.9327 \end{Bmatrix}$

► scale by Σ : $\Sigma V^T e_1 = \begin{Bmatrix} 3.6042 \\ -0.0933 \end{Bmatrix}$, $\Sigma V^T e_2 = \begin{Bmatrix} 1.3935 \\ 0.2414 \end{Bmatrix}$

► rotation again by U : $U\Sigma V^T e_1 = Ae_1 = \begin{Bmatrix} 3 \\ 2 \end{Bmatrix}$, $U\Sigma V^T e_2 = Ae_2 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$

Characteristics of SVD

- ▶ $A^T A = (U \Sigma V^T)^T (U \Sigma V^T) = V(\Sigma^T \Sigma) V^T$,
 $AA^T = (U \Sigma V^T)(U \Sigma V^T)^T = U(\Sigma \Sigma^T) U^T$
- ▶ $AV = U \Sigma U^T V = U \Sigma$, i.e., $A \nu_j = \sigma_j \mu_j$, $j = 1, 2, \dots, n$
- ▶ $A^T U = V \Sigma^T$, i.e., $A^T \mu_j = \sigma_j \nu_j$, $j = 1, 2, \dots, n$ and $A^T \mu_j = 0$,
 $j = n + 1, n + 2, \dots, m$
- ▶ $\sigma_1, \sigma_2, \dots, \sigma_n$ is unique, but U and V^T is not
- ▶ $\text{rank}(A) = \text{rank}(\Sigma) = \mathbb{I}(\sigma_i > 0)$

Matrix Approximation and SVD

$A = [a_{ij}]_{m \times n}$ with $\|A\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n (a_{ij})^2 \right)^{\frac{1}{2}} = \left(\sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2 \right)^{\frac{1}{2}}$, if $\text{rank}(A) = r$ and $A = U\Sigma V^T$, denote $\mathcal{M}$ the set of all matrix in $\mathbb{R}^{m \times n}$ with rank at most k , $0 < k < r$, if X with $\text{rank}(X) = k$ satisfies

$$\|A - X\|_F = \min_{S \in \mathcal{M}} \|A - S\|_F$$

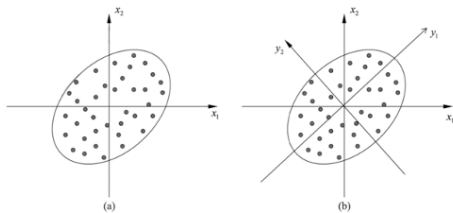
then

$$\|A - X\|_F = \left(\sigma_{k+1}^2 + \sigma_{k+2}^2 + \dots + \sigma_n^2 \right)^{\frac{1}{2}}$$

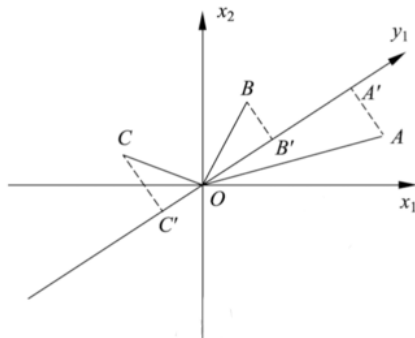
Specifically, if $A' = U\Sigma'V^T$, where $\Sigma' = \begin{Bmatrix} \Sigma_k & 0 \\ 0 & 0 \end{Bmatrix}$, then

$$\|A - A'\|_F = \left(\sigma_{k+1}^2 + \sigma_{k+2}^2 + \dots + \sigma_n^2 \right)^{\frac{1}{2}} = \min_{S \in \mathcal{M}} \|A - S\|_F$$

Principle Component Analysis



Geometry of PCA



Population PCA

- ▶ Suppose $x = (x_1, x_2, \dots, x_m)^\top$ is a m -dim random vector, with
 - ▶ $\mu = E(x) = (\mu_1, \mu_2, \dots, \mu_m)^\top$
 - ▶ $\Sigma = \text{cov}(x, x) = E[(x - \mu)(x - \mu)^\top]$
- ▶ Consider a linear transformation from x to $y = (y_1, y_2, \dots, y_m)^\top$, with $\alpha_i^\top = (\alpha_{1i}, \alpha_{2i}, \dots, \alpha_{mi})$, $i = 1, 2, \dots, m$

$$y_i = \alpha_i^\top x = \alpha_{1i}x_1 + \alpha_{2i}x_2 + \dots + \alpha_{mi}x_m$$

- ▶ Then we know that
 - ▶ $E(y_i) = \alpha_i^\top \mu$
 - ▶ $\text{var}(y_i) = \alpha_i^\top \Sigma \alpha_i$
 - ▶ $\text{cov}(y_i, y_j) = \alpha_i^\top \Sigma \alpha_j$

Population PCA

Given a linear transformation

$$y_i = \alpha_i^\top x = \alpha_{1i}x_1 + \alpha_{2i}x_2 + \dots + \alpha_{mi}x_m$$

if

- ▶ $\alpha_i^\top \alpha_i = 1$
- ▶ $\text{cov}(y_i, y_j) = 0$ for $i \neq j$
- ▶ y_1 has the largest variance among all linear transformations of x ;
 y_2 has the largest variance among all linear transformations that are uncorrelated with $y_1, \dots$

Then, we call $y_1, y_2, \dots, y_m$ the first, second, ..., m -th principle component of x .

Theorem of PCA

Suppose x is m -dim random variable with covariance matrix Σ , $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m \geq 0$ are the eigenvalues of Σ , and $\alpha_1, \alpha_2, \dots, \alpha_m$ are the corresponding eigenvectors, then the k -th principle component of x is given by

$$y_k = \alpha_k^T x = \alpha_{1k}x_1 + \alpha_{2k}x_2 + \dots + \alpha_{mk}x_m$$

for $k = 1, 2, \dots, m$, and the variance of y_k is given by (the k -th eigenvalue of Σ)

$$\text{var}(y_k) = \alpha_k^T \Sigma \alpha_k = \lambda_k$$

Proof: homework?

Theorem of PCA

The components of m -dim random variable $y = (y_1, y_2, \dots, y_m)^T$ are the first, second, ..., m -th principle component of x **iff**

- ▶ $y = A^T x$, where A is orthogonal with $A = \begin{Bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{Bmatrix}$
- ▶ $\text{cov}(y) = \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_m)$, with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$
- ▶ λ_k is the k -th eigenvalue of Σ , α_k is the corresponding eigenvector, for $k = 1, 2, \dots, m$

Characteristics of PCA

- ▶ $\text{cov}(y) = \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_m)$
- ▶ $\sum_{i=1}^m \lambda_i = \sum_{i=1}^m \sigma_{ii}$, where σ_{ii} is the variance of x_i
- ▶ factor loading: correlation coefficient between y_k and x_i ,
$$\rho(y_k, x_i) = \frac{\sqrt{\lambda_k} \alpha_{ik}}{\sqrt{\sigma_{ii}}}$$
- ▶ sum of factor loading for y_k over x : $\sum_{i=1}^m \sigma_{ii} \rho^2(y_k, x_i) = \lambda_k$
- ▶ sum of factor loading for x_i over y : $\sum_{k=1}^m \rho^2(y_k, x_i) = 1$

Practicalities of PCA

How many principle components to choose?

- ▶ variance contribution of y_k : $\eta_k = \frac{\lambda_k}{\sum_{i=1}^m \lambda_i}$
- ▶ cumulative variance contribution of $y_1, y_2, \dots, y_k$: $\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^m \lambda_i}$
- ▶ usually choose k such that $\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^m \lambda_i} > 70\%$ or 80%
- ▶ cumulative variance contribution reflects the contribution of $y_1, \dots, y_k$ to x , but not specifically to x_i
- ▶ contribution to x_i : $\nu_i = \rho^2(x_i, (y_1, \dots, y_k)) = \sum_{j=1}^k \frac{\lambda_j \alpha_{ij}^2}{\sigma_{ii}}$

Note: normalize variables before PCA

Sample PCA

Suppose m -dim random variable x is repeatedly measured n times, giving sample data $x_1, x_2, \dots, x_n$, where $x_j = (x_{1j}, x_{2j}, \dots, x_{mj})^T$ indicates the j -th measurement, and x_{ij} indicates the j -th measurement of the i -th variable, for $j = 1, 2, \dots, n$ and $i = 1, 2, \dots, m$. Denote sample data matrix as

$$X = [x_1, x_2, \dots, x_n] = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \dots & \dots & \dots & \dots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix}$$

PCA performed on the sample matrix is called sample PCA and inherent the same characteristics of the population PCA.

PCA Toy Example

n students take exams for 4 courses. Below is the correlation matrix for exam scores

course	Chinese	English	Math	Physics
Chinese	1	0.44	0.29	0.33
English	0.44	1	0.35	0.32
Math	0.29	0.35	1	0.6
Physics	0.33	0.32	0.6	1

Perform PCA for the sample data.

PCA Toy Example

- ▶ Denote x_1, x_2, x_3, x_4 as the exam score for Chinese, English, Math and Physics.
- ▶ Perform eigen decomposition to the correlation matrix
 - ▶ eigenvalues: $\lambda_1 = 2.17, \lambda_2 = 0.87, \lambda_3 = 0.57, \lambda_4 = 0.39$
 - ▶ variance contribution $\frac{\lambda_1 + \lambda_2}{\sum_{i=1}^4 \lambda_i} = 0.76$
 - ▶ seems that $k = 2$ is sufficient
 - ▶ corresponding eigenvector

PC	x_1	x_2	x_3	x_4	variance contribution
y_1	0.460	0.476	0.523	0.537	0.543
y_2	0.574	0.486	-0.476	-0.456	0.218

PCA Toy Example

So the first and second PC are

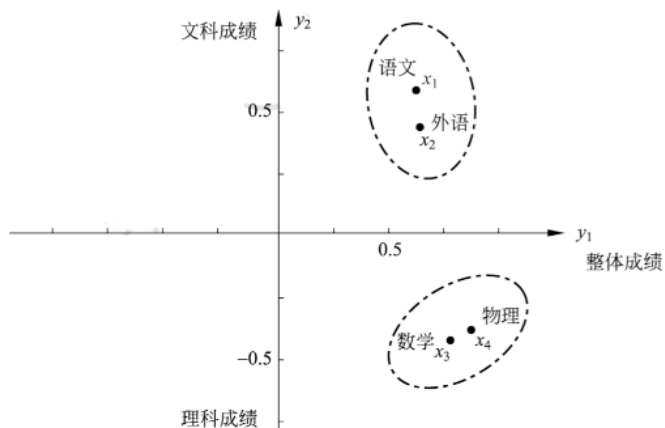
$$y_1 = 0.460x_1 + 0.476x_2 + 0.523x_3 + 0.537x_4$$

$$y_2 = 0.574x_1 + 0.486x_2 - 0.476x_3 - 0.456x_4$$

Factor loading and contribution to x_i , $i = 1, 2, 3, 4$

PC	x_1	x_2	x_3	x_4
y_1	0.678	0.701	0.770	0.791
y_2	0.536	0.453	-0.444	-0.425
joint contribution to x_i	0.747	0.697	0.790	0.806

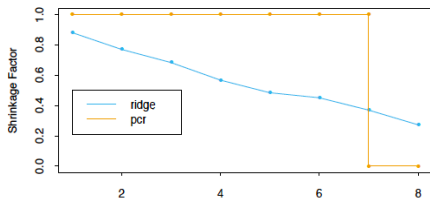
PCA Toy Example



Principle Components Regression

Eigen decomposition of $X^T X$: $X^T X = V D^2 V^T$

- ▶ Eigen vector v_j (column of V): principal components directions
- ▶ $z_j = X v_j$: principle components
- ▶ z_1 has the largest sample variance amongst all normalized linear combinations of the columns of X .
- ▶ Derived input columns $(z_1, z_2, \dots, z_M)$, where $M \leq p$
- ▶ $\hat{y}_{(M)}^{pcr} = \bar{y}\mathbb{I} + \sum_{m=1}^M \hat{\theta}_m z_m$



Partial Least Squares

Use the output Y (in addition to X) in construction of the derived input

- ▶ Compute $\hat{\phi}_{1j} = \langle x_j, y \rangle$ for each $j = 1, \dots, p$
- ▶ $z_1 = \sum_j \hat{\phi}_{1j} x_j$: first partial least squares direction
- ▶ Output y is regressed on z_1 giving coefficient $\hat{\theta}_1$
- ▶ Orthogonalize $x_1, \dots, x_p$ w.r.t. z_1
- ▶ Continue until $M \leq p$ directions have been obtained

Note: Each z_m is weighted by the strength of their univariate effect on y

Partial Least Squares - Algorithm

- ▶ Standardize each x_j to have mean zero and variance one. Set $\hat{y}^{(0)} = \bar{y}$, $x_j^{(0)} = x_j$, $j = 1, \dots, p$.
- ▶ For $m = 1, 2, \dots, p$
 1. $z_m = \sum_{j=1}^p \hat{\phi}_{mj} x_j^{(m-1)}$, where $\hat{\phi}_{mj} = \langle x_j^{(m-1)}, y \rangle$
 2. $\hat{\theta}_m = \frac{z_{m,y}}{z_{m,z_m}}$
 3. $\hat{y}^{(m)} = \hat{y}^{(m-1)} + \hat{\theta}_m z_m$
 4. Orthogonalize each $x_j^{(m-1)}$ w.r.t. z_m : $x_j^{(m)} = x_j^{(m-1)} \frac{z_{m,x_j^{(m-1)}}}{z_{m,z_m}} z_m$, $j = 1, \dots, p$
- ▶ Output the sequence of fitted vectors $\{\hat{y}^{(m)}\}$, recover coef $\hat{y}^{(m)} = X \hat{\beta}^{PLS}(m)$

Comparison - PCR & PLS

PCR: seeks directions that have high variance

- ▶ The m_{th} principal component direction ν_m solves
- ▶ $\max_{\alpha} \text{Var}(X\alpha)$
- ▶ subject to $\|\alpha\| = 1, \alpha^T S \nu_l = 0, l = 1, \dots, m-1$

PLS: seeks directions that have high variance & high correlation with the output

- ▶ The m_{th} PLS direction $\hat{\phi}_m$ solves
- ▶ $\max_{\alpha} \text{Corr}^2(y, X\alpha) \text{Var}(X\alpha)$
- ▶ subject to $\|\alpha\| = 1, \alpha^T S \hat{\phi}_l = 0, l = 1, \dots, m-1$

Note: the variance aspect tends to dominate, so PLS behaves much like ridge and PCR

Open Discussion

Compare subset selection, shrinkage methods, derived input methods

Pros & Cons?

In terms of sample variance, prediction error, smooth vs discrete, computational tractability?